

StockSense AI

Retail sales forecasting and stock planning

VUT AI Solutions

Business Analysis 3.2 AIBUY3A

The retail ordering problem

Manual ordering lacks a consistent estimate for the period ahead.

Too little stock can create shortages. Too much can tie up money.

The manager keeps control of the ordering decision.

Business objectives and scope

Seven daily forecasts for store 1, items 1-5

A stock suggestion with visible assumptions

One local page with a short text assistant

Real inventory benefits need a business trial.

The sales dataset

913,000 source rows from 2013-2017

9,130 selected rows with complete daily histories

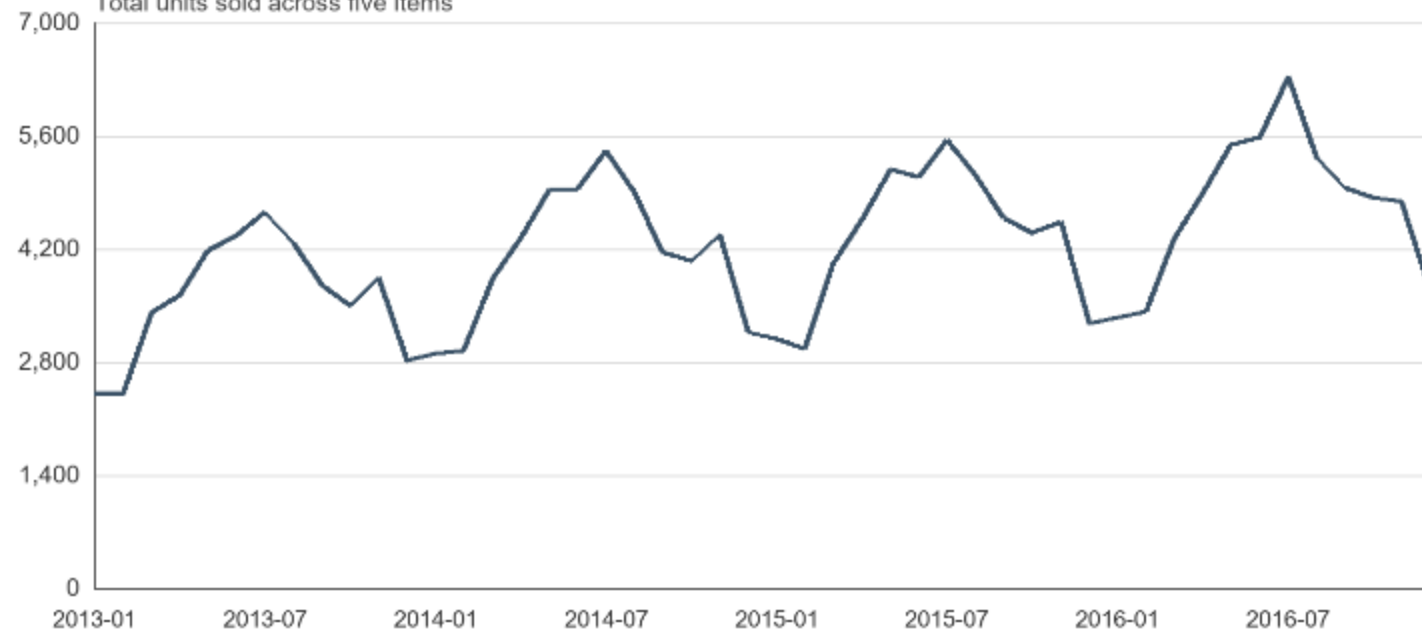
Fields: date, store, item and sales

Anonymous item IDs and manually entered stock inputs

Sales patterns in development data

Monthly sales from 2013 to 2016

Total units sold across five items



Source: supplied train.csv. Store 1, items 1 to 5. Development data only.

Annual total
43,208 in 2013
56,943 in 2016

Forecasting methods and fair evaluation

Weekly benchmark repeats matching weekdays.

Random forest uses sales lags, recent means and calendar inputs.

Small LSTM uses chronological sales sequences.

Fit 2013-2015, select on 2016, refit through 2016, test 2017.

2016 validation selected random forest

Method	Daily MAE	Daily RMSE	Weekly-total MAE
Weekly benchmark	6.320	8.217	18.758
Random forest	4.911	6.596	19.797
Small LSTM	5.439	7.227	17.280

Errors in units. The method choice stayed fixed before 2017 scoring.

2017 final test

Method	Daily MAE	Daily RMSE	Weekly-total MAE
Weekly benchmark	6.273	8.218	18.427
Random forest	4.812	6.387	17.973
Small LSTM	5.411	7.261	17.343

Random forest lowers daily MAE by about 23.3%.
LSTM has the lowest overall weekly-total error.

The small LSTM experiment

28 sales inputs, 16 LSTM units, seven outputs

1,271 trainable parameters per item

Training-only scaling and validation early stopping

Frozen refit epochs: 39, 50, 12, 50 and 21

Stock planning

Coverage = lead time + review interval, up to seven days

Target = forecast over coverage + buffer

Order = round upwards $\max(0, \text{target} - \text{stock} - \text{outstanding})$

A shortage warning checks sales before delivery.

NLP and the text assistant

TF-IDF word features and logistic regression recognise five intents.

Templates retrieve the same forecast and stock objects shown on screen.

Fresh check: 24/25 supported requests, 8/8 unrelated requests rejected.

Personal PC verification

45 automated tests passed with no skips

All five items, stock rules and assistant consistency checked

AppTest initial render 7.29 seconds

Warm selections 0.20-0.23 seconds

Application demonstration

Select an item and inspect seven dated estimates.

Calculate an order, then show invalid coverage and a shortage.

Ask for the forecast and open both historical evaluation tables.

Findings and remaining evidence

The local prototype works and random forest meets the daily error criterion.

LSTM shows a weekly-total trade-off worth investigating.

Human review, signed declarations and Grammarly evidence remain.

Questions